

PRACTICAL 1

TOPIC: DDL, DML, DCL, TCL COMMANDS

SECTION 1 (DDL)

```
mysql> CREATE DATABASE bankdb;
Query OK, 1 row affected (0.01 sec)

mysql> USE bankdb;
Database changed
mysql> CREATE TABLE bank_transactions (
  ->     txn_id INT PRIMARY KEY,
  ->     customer_name VARCHAR(50),
  ->     branch_name VARCHAR(50),
  ->     transaction_type VARCHAR(20),
  ->     amount DECIMAL(10,2),
  ->     transaction_date DATE
  -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> SHOW TABLES;
+-----+
| Tables_in_bankdb |
+-----+
| bank_transactions |
+-----+
1 row in set (0.00 sec)
```

```
mysql> ALTER TABLE bank_transactions
  -> ADD account_no VARCHAR(20);
Query OK, 0 rows affected (0.11 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> DESCRIBE bank_transactions;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| txn_id         | int           | NO   | PRI | NULL    |       |
| customer_name  | varchar(50)   | YES  |     | NULL    |       |
| branch_name    | varchar(50)   | YES  |     | NULL    |       |
| transaction_type | varchar(20)   | YES  |     | NULL    |       |
| amount         | decimal(10,2) | YES  |     | NULL    |       |
| transaction_date | date          | YES  |     | NULL    |       |
| account_no     | varchar(20)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.01 sec)
```

```
mysql> ALTER TABLE bank_transactions
-> MODIFY customer_name VARCHAR(100);
Query OK, 0 rows affected (0.11 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESCRIBE bank_transactions;
```

Field	Type	Null	Key	Default	Extra
txn_id	int	NO	PRI	NULL	
customer_name	varchar(100)	YES		NULL	
branch_name	varchar(50)	YES		NULL	
transaction_type	varchar(20)	YES		NULL	
amount	decimal(10,2)	YES		NULL	
transaction_date	date	YES		NULL	
account_no	varchar(20)	YES		NULL	

```
7 rows in set (0.01 sec)
```

```
mysql> RENAME TABLE bank_transactions
-> TO customer_transactions;
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> SHOW TABLES;
```

Tables_in_bankdb
customer_transactions

```
1 row in set (0.00 sec)
```

```
mysql> TRUNCATE TABLE customer_transactions;
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> SHOW TABLES;
```

Tables_in_bankdb
customer_transactions

```
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM customer_transactions;
Empty set (0.00 sec)
```

```
mysql> DROP TABLE bank_backup;
ERROR 1051 (42S02): Unknown table 'bankdb.bank_backup'
mysql> CREATE TABLE bank_backup (
    ->     txn_id INT PRIMARY KEY,
    ->     customer_name VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> DROP TABLE bank_backup;
Query OK, 0 rows affected (0.03 sec)
```

SECTION 2 (DML)

```
mysql> INSERT INTO customer_transactions
    -> (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
    -> VALUES
    -> (101, 'Ravi', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
    -> (102, 'Sita', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
    -> (103, 'Kiran', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
    -> (104, 'Anil', 'Vizag', 'Deposit', 8000, '2024-01-10'),
    -> (105, 'Priya', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
    -> (106, 'Ramesh', 'Vizag', 'Deposit', 15000, '2024-01-12'),
    -> (107, 'Keerthi', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
    -> (108, 'Rahul', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
    -> (109, 'Sneha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
    -> (110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
Query OK, 10 rows affected (0.02 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO customer_transactions
    -> (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
    -> VALUES
    -> (111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT * FROM customer_transactions;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Priya	Hyderabad	Withdrawal	3500.00	2024-01-11	NULL
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL
111	Venu	Vizag	Deposit	7000.00	2024-01-18	NULL

```
11 rows in set (0.00 sec)
```

```
mysql> UPDATE customer_transactions
-> SET amount = 5000
-> WHERE txn_id = 105;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM customer_transactions
-> WHERE txn_id = 105;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> RENAME TABLE customer_transactions TO bank_transactions;
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> DELETE FROM customer_transactions
-> WHERE txn_id = 111;
ERROR 1146 (42S02): Table 'bankdb.customer_transactions' doesn't exist
mysql> DELETE FROM bank_transactions
-> WHERE txn_id = 111;
Query OK, 1 row affected (0.02 sec)

mysql> SELECT * FROM bank_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 5000.00 | 2024-01-05 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 | NULL |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | NULL |
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 | NULL |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00 | 2024-01-13 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 | 2024-01-14 | NULL |
| 109 | Sneha | Vizag | Withdrawal | 4000.00 | 2024-01-15 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 | 2024-01-16 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM bank_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 5000.00 | 2024-01-05 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 | NULL |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | NULL |
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 | NULL |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00 | 2024-01-13 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 | 2024-01-14 | NULL |
| 109 | Sneha | Vizag | Withdrawal | 4000.00 | 2024-01-15 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 | 2024-01-16 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.01 sec)

mysql> SELECT COUNT(*) FROM bank_transactions;
+-----+
| COUNT(*) |
+-----+
| 10 |
+-----+
1 row in set (0.01 sec)
```

```
mysql> SELECT *
-> FROM bank_transactions
-> WHERE transaction_type = 'Deposit';
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 5000.00 | 2024-01-05 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 | NULL |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 | NULL |
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 | 2024-01-14 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 | 2024-01-16 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM bank_transactions
-> ORDER BY amount DESC;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL

10 rows in set (0.00 sec)

SECTION 3 : DCL

```
mysql> CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.04 sec)

mysql> GRANT SELECT
-> ON bank_transactions
-> TO 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'Auditor1'@'localhost';
```

Grants for Auditor1@localhost
GRANT USAGE ON *.* TO 'Auditor1'@'localhost'
GRANT SELECT ON `bankdb`.`bank_transactions` TO 'Auditor1'@'localhost'

2 rows in set (0.00 sec)

```
mysql> CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'manager123';
Query OK, 0 rows affected (0.02 sec)

mysql> GRANT ALL PRIVILEGES
-> ON bank_transactions
-> TO 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'BranchManager'@'localhost';
```

Grants for BranchManager@localhost
GRANT USAGE ON *.* TO 'BranchManager'@'localhost'
GRANT ALL PRIVILEGES ON `bankdb`.`bank_transactions` TO 'BranchManager'@'localhost'

2 rows in set (0.00 sec)

```
mysql> REVOKE SELECT
-> ON bank_transactions
-> FROM 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'Auditor1'@'localhost';
```

Grants for Auditor1@localhost
GRANT USAGE ON *.* TO 'Auditor1'@'localhost'

1 row in set (0.00 sec)

```
mysql> REVOKE ALL PRIVILEGES
      -> ON bank_transactions
      -> FROM 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'BranchManager'@'localhost';
+-----+
| Grants for BranchManager@localhost |
+-----+
| GRANT USAGE ON *.* TO 'BranchManager'@'localhost' |
+-----+
1 row in set (0.00 sec)
```

SECTION 4 : TCL

```
mysql> UPDATE bank_transactions
      -> SET amount = 6000
      -> WHERE txn_id = 101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT *
      -> FROM bank_transactions
      -> WHERE txn_id = 101;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 6000.00 | 2024-01-05 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> |
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE bank_transactions
      -> SET amount = 8000
      -> WHERE txn_id = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> ROLLBACK TO Before_Update;
Query OK, 0 rows affected (0.01 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE bank_transactions
    -> SET amount = 99999
    -> WHERE txn_id = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> ROLLBACK TO Before_Update;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT *
    -> FROM bank_transactions
    -> WHERE txn_id = 102;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> |
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE bank_transactions
    -> SET amount = 7000
    -> WHERE txn_id = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SAVEPOINT SP1;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE bank_transactions
    -> SET amount = 9000
    -> WHERE txn_id = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> ROLLBACK TO SP1;
Query OK, 0 rows affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT *
    -> FROM bank_transactions
    -> WHERE txn_id IN (101,102);
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 7000.00 | 2024-01-05 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> |
```

